Unemployment Benefits, Food Insecurity, and Supplemental Nutrition Assistance Program Spending

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This paper examines the effects of Unemployment Insurance (UI) expansions on household food insecurity in the United States using the Food Security Supplement to the Current Population Survey for 1999-2017. We exploit the plausibly exogenous variation in state maximum weekly UI benefits across states and over time. The two-part model demonstrates that a $100 increase in state weekly UI benefits reduces the likelihood of food insecurity among UI-eligible families by 5.6% and the severity of food insecurity conditional on being food insecure by 6.3%. We further present strong evidence of program substitution between UI and the Supplemental Nutrition Assistance Program (SNAP), which is the largest food assistance program with the stated goal of reducing food insecurity. Specifically, for every dollar spent on UI, 4-11 cents can be saved on SNAP, thereby reinforcing the social benefits of an UI expansion.

KEYWORDS: food insecurity, SNAP, unemployment insurance.

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During the Covid-19 pandemic in 2020 when the unemployment rates in the United States sharply increased, household food insecurity has also worsened. In hope to provide at least temporary financial supports for families in need, unemployment insurance (UI) programs have been expanded at both the federal and state levels to offer more benefits, extended durations of claiming UI benefits, and broader coverages to all individuals who are out of work due to the pandemic (Raifman, Bor, and Venkataramani 2020). Would an UI expansion as such be effective in improving food security among affected households? This is in fact an empirical question even outside the pandemic period. In this paper, we examine the causal effects of UI expansions on the likelihood of food insecurity and the conditional severity of food shortages among the UI-eligible households.

The extensive literature evaluating social insurance programs, such as UI, has typically concluded that generous insurance has clear social benefits of consumption smoothing (e.g., Gruber 1997).[1] Therefore, more generous UI is likely to improve household food intake through consumption smoothing. However, estimates of the consumption-smoothing benefits are often small. For example, Browning and Crossley (2001) find that a 10-percentage point increase in the UI generosity leads to a 0.8% reduction in the consumption drop upon job loss. To this end, the extent to which UI affects household food insecurity might also be small. But it is also possible that the benefits of consumption smoothing are nonlinear in a way such that the consumption smoothing that prevents food insufficiency is substantial. Overall, the size of the UI expansion effect on household food insecurity remains unknown.

Furthermore, since most food programs are means-tested, an increase in UI benefits is likely to reduce reliance on food assistance programs, such as the Supplemental Nutrition Assistance Program (SNAP). A shift from food to cash benefits might crowd out food spending and might not

necessarily improve household food security. In addition, Campos and Reggio (2020) open up the possibility that, in reaction to earnings shocks, low-income households may change their food consumption through food prices rather than the quantities of food they consume.

This paper fills the gap in the literature by providing a direct evaluation of how a more generous UI affects household food insecurity using the Current Population Survey-Food Security Supplement (CPS-FSS) between 1999 and 2017. We adopt the two-part model to account for the mixed distribution that contains numerous zeros in the dependent variable. Specifically, in the first part, we define food-insecure households as those reporting some hardship in any of the 18 questions related to their food needs. In the second part, which is conditional on being food insecure, households are further characterized using Rasch scores, which reflect the severity of food insecurity (Gundersen 2008).[2]

To address concerns that variation in actual UI benefits received by participating families is likely endogenous due to strong self-selection, we exploit the plausibly exogenous variation in state maximum weekly UI benefits (i.e., the largest possible benefits a household can receive). They are determined by state UI laws; therefore, they are uncorrelated with household characteristics in principle. Since changes in state maximum weekly UI benefits are not necessarily binding for all UI-recipients, this identification strategy estimates the intention-to-treat rather than the average treatment effect. A similar approach has been used to analyze the effects of unemployment benefits on job search behaviors (Krueger and Mueller 2010), health conditions (Kuka 2020), academic achievements among children (Regmi 2018), and cigarette smoking (Fu and Liu 2019).

We find that a $100 (approximately 20%) increase in maximum weekly UI benefits reduces the likelihood of being food insecure by about 2.3 percentage points for families with unemployed

members, excluding re-/new entrants. When we further limit the sample to those who reported having experienced some food shortages, a $100 increase in maximum weekly UI benefits leads to an approximate 6.3% reduction in the severity of food insufficiency. By adopting the March CPS, which contains information on UI income, we identified a positive income effect resulting from increases in the unemployment benefits as a potential mechanism that reduces food insecurity.

The benefits of improving food security should be weighed against the expenditure changes for related food programs, namely SNAP, whose stated goal is to reduce food insecurity among low-income households. If UI and SNAP are substitutes for each other, then the social benefits of UI expansions should be larger after accounting for the costs saved on SNAP. By contrast, if the two programs are complements, this benefit is likely offset by the extra SNAP spending.

By further exploring information in the March CPS on SNAP enrollment status and the amount of SNAP benefits received, we reveal program substitution in which an increase in unemployment benefits reduces spending on SNAP. Specifically, a $100 increase in maximum weekly UI benefits reduces the likelihood of SNAP participation by around 7.1%. Such an expansion in UI also decreases the annual amount of claimed SNAP benefits by about $178, or equivalently 6.2%, when the analysis is further restricted to SNAP enrollees. A welfare analysis suggests that for every additional dollar spent on the UI program, SNAP expenditures decrease by approximately 4-11 cents.

This paper is relevant to a broad range of literature. First, our work is directly related to the literature on social programs and food insecurity. Existing studies have mainly focused on the effects of food assistance programs (e.g., Borjas 2004; Nord and Prell 2011), whereas research on how nonfood programs affect household food insecurity is relatively limited. The few exceptions,

mostly published by the public health community, studied the Earned Income Tax Credit (Heflin 2014), minimum wages (Sabia and Nielsen 2015), Temporary Assistance for Needy Families (Lee, Slack, and Lewis 2004), and a combined measure of various social welfare programs (Schmidt, Shore-Sheppard, and Watson 2016). Our study adds to this strand of literature by focusing on the UI program. We utilize variation in state unemployment benefit levels rather than UI duration (Bartfeld and Men 2017) because the latter has remained stable apart from the Great Recession period (Fu and Liu 2019).

Second, this paper adds to the literature that has examined interactions between nonfood safety net programs and food assistance programs, namely SNAP. Existing studies have found both positive program interactions possibly due to an information effect (e.g., Schmidt, Shore-Sheppard, and Watson 2019) and negative program interactions resulting from higher incomes (e.g., Reich and West 2015). Leung and O'Leary (2020) also examine the interactions between UI and other welfare programs, including SNAP. They study on the margin of UI eligibility, while our paper focuses on a sample of households that qualify for UI. The disparity between our results and the results therein reveals that program interactions differ between eligible and marginally eligible households.

Third, relevant studies have typically ignored the mixed distribution of household food insecurity and used binary or categorical measures (e.g., Heflin 2014; Schmidt, Shore-Sheppard, and Watson 2016; Bartfeld and Men 2017). The mixed distribution arises because many households do not experience food insecurity, whereas for those that do, the distribution of the intensity of food insecurity is approximately continuous. Such a mixed distribution of the outcome variable, which is commonly observed in studies on health care demand (e.g., Manning et al. 1987) and cigarette consumption (e.g., Wasserman et al. 1991; Tauras 2005; Kostova et al. 2011),

motivates the adoption of the two-part model.[3] The model also enables us to disentangle two marginal effects, namely the effect of increasing unemployment benefits on the likelihood of food insecurity and on the severity of food insecurity, conditional on being food insecure.

Finally, our work is related to the literature on designing an optimal UI program. Our estimates confirm the benefits of UI. Consumption smoothing is even more valuable than previously assumed because of the effect it has on improving food sufficiency. Furthermore, our results reconcile with the recent literature that has accounted for interactions with other safety net programs in calculating the optimal level of UI provision (e.g., Lawson 2015). The findings that increased UI reduces SNAP spending provide additional evidence of program substitution. Although we do not attempt to impute the optimal level of UI benefits in this paper, our estimates reemphasize the role of accounting for interaction with other programs in optimal UI design.

The paper is organized as follows: Section 2 explains the institutional backgrounds on both UI and SNAP; Section 3 describes the data used in our analyses; Section 4 presents the estimation model and our primary results with a series of robustness checks; Section 5 discusses the mechanism and performs a cost-benefit analysis of program substitution; finally, Section 6 concludes.

Institutional Background

UI

UI is a joint federal-state program mainly funded by federal and state payroll taxes levied on employers. It provides temporary financial assistance to individuals who have experienced job loss

through no fault of their own. All states comply with the guidelines established by federal law, but they may design their own UI programs. To be eligible for unemployment benefits, individuals typically must have worked for at least one quarter of the past year and have been involuntarily laid off by their employer. They must also meet state requirements for earnings. In general, the standard maximum duration of unemployment compensation is 26 weeks, which is often extended during economic downturns. Benefits are typically set as a percentage of an individual's weekly earnings prior to unemployment, up to a fixed percentage of the state's average weekly wages. Wage differences among states generate substantial variation in state maximum weekly UI benefits. For example, in 2011, the maximum weekly UI benefits were $625 in Massachusetts, whereas in Arizona, they were $240. Additional variation in UI policies has occurred over time as states have updated their benefit schedules periodically to match wage changes.[4]

SNAP

SNAP, formerly known as the Food Stamp Program, is a federal aid program administered by the USDA. It aims to improve food security by providing food-purchasing assistance to low-income families. Enrollment in SNAP and spending of SNAP benefits are countercyclical. SNAP caseloads increased significantly during the Great Recession, peaked in 2013, and began to decrease after 2014 as the economy recovered (Rosenbaum and Keith-Jennings 2016). Eligibility for the program is universal. Households must earn a gross monthly income of no more than 130% of the federal poverty level for a specific household size to be eligible for SNAP benefits. Benefits are distributed using an electronic benefit transfer card each month and can be used to purchase only food. Households are expected to spend approximately 30% of their incomes on food; thus,

SNAP benefits are calculated as the difference between 30% of a household's net monthly income and the maximum monthly benefits designated on the basis of household size.

The relationship between UI and SNAP is indistinct. An increase in UI benefits might mechanically reduce SNAP assistance because of increased household income. However, given the widely reported positive effect of UI generosity on unemployment duration in the literature, if longer unemployment leads to more SNAP applications, then we would expect UI and SNAP to have a positive connection. Moreover, the two programs target different populations. If many UI recipients already have incomes above the federal poverty level and are thus ineligible for SNAP, UI expansion may have a small effect on SNAP participation and expenditure. Similarly, SNAP recipients who are employed or have exhausted their UI benefits before obtaining new jobs, are unlikely to be affected by increased UI benefits.

Data

Our main analyses are based on two data sources: the CPS-FSS from 1999 to 2017 and March CPS over the same period. We use the CPS-FSS to examine the UI expansion impacts on household food insecurity; and we separately study the program interconnectedness between UI and SNAP based on March CPS. We do not limit our sample to the Outgoing Rotation Group of the CPS; and hence, the data used in both analyses are repeated cross-sectional.

To construct a sample of UI-eligible households, we use the CPS survey question that asks the unemployed individuals about their reasons for unemployment, and summarize the answers into four categories, namely job losers (due to temporary layoff, involuntary job loss, or ending of a temporary job), job leavers, re-entrants who were re-entering the labor market after an extended

absence from the work force, and new entrants who were seeking their first jobs. We make two restrictions to arrive at our main analysis sample. First, we restrict our baseline sample to families with at least one member who was unemployed at any time during the past 26 weeks. We use 26 weeks as the reference period because most states had a maximum duration of 26 weeks for the receipt of unemployment benefits during our sample period, apart from the Great Recession. Second, a UI claimer must have sufficient earnings in the base period (often four quarters preceding the claim) in order to be eligible for the benefits (Leung and O'Leary 2020). Therefore, we exclude the re-/new entrants because they are unlikely to have positive earnings in base periods.[5] We later test the robustness of our baseline results to a battery of alternative sample choices in Section IV. Throughout the analyses, we exclude households with no members aged 20-60 years.

Household-level food security data were collected from the CPS-FSS from 1999 to 2017. Food insecurity status is defined on the basis of the 18 survey questions on conditions of food shortages within households.[6] These questions are asked to households with incomes less than 185% of the federal poverty level or higher-income families who report food shortages in the two screening questions. In each of the 18 questions, households are asked whether a specific food hardship occurred at any time during the past 12 months because of a lack of money. For example, households are asked whether they are worried that their food will run out before they receive money to buy more and whether the adults in the family reduced the size of or skipped meals in ≥3 months. Food insecurity status in the survey is then determined using the number of affirmative responses. Households are classified as being food secure if they report no shortages in any of the 18 questions. Households have moderate food security if they report one or two affirmative answers, low food security if they report three to seven affirmative responses, and very low food

security if they report eight or more affirmative answers. Relevant studies have often adopted this measure or categorized households reporting high or marginal food security as being food sufficient and households with low or very low food security as being food insufficient.

The USDA further employs a Rasch measurement model to account for differences in food insecurity severity among questions. The Rasch model is an item response theory model that uses sophisticated statistical imputation to summarize multiple binary answers.[7] In the Rasch model, different values are assigned to different numbers of positive answers, and more affirmative responses are associated with higher scores, suggesting higher levels of food insecurity.[8]

We explore two outcomes: the propensity to be food insecure and the severity of food insecurity conditional on being food insecure. First, food insecurity status is coded as 1 if the household reports affirmative answers to at least one of the 18 food insecurity questions and 0 if no food-insecure conditions are reported.[9] We construct the extensive margin of household food insecurity in this way to account for its mixed distribution that contains numerous zeros. This is particularly important for the UI-eligible population since many of them are ineligible for SNAP (see Table 1) and are much less likely than SNAP benefit recipients to suffer from severe food shortages. If we consider households who report no difficult conditions among all 18 situations as being fully food secure, this binary outcome variable allows us to capture any movement away from full food security, the margin where most UI-eligible households are likely located. We later expand this definition by studying whether the UI impacts are mostly driven by transitions of households with severe food insufficiency or those who are moderately food insecure in Section V.

Next, we use the Rasch food security score provided in the survey to define the severity of food-insecure households. We prefer the Rasch score to the commonly used four-category variable for two reasons. First, the Rasch scores measure the severity of food insecurity more precisely. For

example, households who respond affirmatively to eight questions are treated the same as those who answer affirmatively to nine questions in the four-category variable, but they are assigned different scores of 6.61 and 7.18, respectively. Second, the Rasch scores can reflect the concavity of household food insecurity. For example, the Rasch score increases by 0.73 when the number of positive answers increases three to four, whereas the score increases by only 0.57 when the number of affirmative responses increases from eight to nine.

As presented in the first two columns of Table 1, according to our definition of food insecurity, 42.1% of the UI-eligible households have experienced some food difficulty. Among them, the average Rasch score is approximately 4.77, which is equivalent to about 5 affirmative responses out of the 18 questions. By contrast, the commonly used binary measure for household food insecurity status reduces the share of the food insecure UI-eligible families to 28%, which is close to what has been reported in the literature that uses the same measure. For example, in Schmidt, Shore-Sheppard, and Watson (2016), 33% of the households under study are food insecure.

To study the interconnection between UI and SNAP, we separately explore the March CPS data because it contains information on household SNAP recipiency. Specifically, the March CPS asks households with estimated incomes below a given level if any member received benefits from SNAP during the past year. The survey also obtains the total value of SNAP benefits received by each household. For the analyses of program interconnectedness, we restrict our sample to actual UI recipients reporting positive unemployment benefits.[10] We exclude individuals who report a topcoded value of unemployment benefits. These individuals account for approximately 1% of all SNAP recipients each year.[11] As a robustness check, we later exclude households whose average unemployment benefits per UI recipient in the family are less than $100 per year to account for reporting errors. Our sample is not limited to the subgroup of UI recipients who are eligible for

SNAP because the sample would be stratified on an endogenous variable if UI expansion affects SNAP eligibility perhaps through an income effect. Summary statistics are reported in the last two columns of Table 1. On average, participating families receive approximately $3,964 in annual unemployment benefits. Among the UI recipients, around 13.7% are also enrolled in SNAP, and they claim an average of approximately $2,784 in SNAP benefits per year.

As mentioned, in concern of potential selection bias, we do not use the actual UI benefits received by each family. Instead, we exploit the variation in the policy parameters of state UI laws. We collect both the maximum weekly UI benefits and the maximum benefit durations for all US states from the US Department of Labor's Comparison of State Unemployment Laws report between 1999 and 2017.[12] States with more UI benefits are likely to be more generous with other welfare programs; this relationship might affect household food insecurity. Therefore, we control for a wide range of state-level SNAP policy instruments and a series of local welfare policies. Regarding the state-level SNAP policy environments, we include SNAP benefits for two-person families, SNAP participation rates, SNAP taking-up stigma index, number of call centers for SNAP,[13] a dichotomous variable for whether state offers transitional SNAP benefits to families leaving the TANF or state-funded cash assistance programs, indicators for whether state eliminates the asset test under broad-based categorical eligibility,[14] and a set of variables for the proportions of elderly SNAP units with 1-3 months' recertification periods, 4-6 months' recertification periods, and 7-12 months' certification periods.

To eliminate the potential impacts from other local welfare policies, we control for Earned Income Tax Credit (EITC) benefits for families with two dependents, participation rates for welfare programs including the School Breakfast Program, the National School Lunch Program, Medicaid, and the Special Supplemental Nutrition Program for Women, Infants, and Children

(WIC), benefits of Aid to Families with Dependent Children (AFDC) for 2-person households, and state EITC refund status. We further include state-level unemployment rates, poverty rates, union coverage, minimum wages, the logarithm of food and beverages purchased for off-premises consumption per capita, and the logarithm of state gross domestic product per capita to control for local economic conditions, and the democratic status of the state governor and the logarithm of state population size to absorb the potential impacts from state-level demographic characteristics. The SNAP policy variables are drawn from the U.S. Department of Agriculture SNAP Policy Datasets, and food and beverages purchased for off-premises consumption per capita come from the ArchivaL Federal Reserve Economic Data (AFRED). All other state-level policy variables and economic and demographic statistics are from the University of Kentucky Center for Poverty Research (UKCPR) National Welfare Database (UKCPR 2016). We convert all dollar values to 2018 dollars by using the Urban Consumer Price Index (CPI-U) for food.

Finally, household characteristics that are typically associated with food insecurity are included. Specifically, we control for the age and marital status of the head of household; the share of white household members; the share of members with less than a high school education, a high school degree, or some college education or above; the share of members aged 20-60 years; indicators for different household sizes (one-person households and families with two to five members), and an indicator for the presence of children younger than 6 years. The descriptive statistics for these control variables are provided in the lower panel of Table 1.

Estimation Strategy and Results

Food Insecurity

We use the two-part model to reflect the prevalence of zero food insecurity in our sample, as opposed to the sample selection model explained in footnote 2. In the first part of the model, we express the likelihood of household-level food insecurity (*y*) as a linear function of state maximum weekly UI benefits (*B*), state maximum benefit durations (*R*), a vector of household characteristics (*X*), and a series of state-level controls (*Z*) that include state economic attributes and complementary welfare policies and programs. We also include state fixed effects (δ_s) and year fixed effects (τ_t). To allow the time trend effects to vary across states, we further control for state-specific linear time trends ($\eta_s t$). As a robustness check, results are reported with and without this term. Standard errors are clustered at the state level to allow for possible within-state correlations between households. We use the CPS-FSS weights and express the maximum weekly UI benefits (B) in hundreds for all analyses.

The first part of the model is specified as follows:

$$(1) \qquad \Pr(y_{ist}) = \beta_0 + \beta_1 B_{st} + \beta_2 R_{st} + X_{ist}\delta_1 + Z_{st}\delta_2 + (\eta_s t) + \tau_t + \delta_s$$

where *i* denotes household, *s* represents state, and *t* stands for year. We estimate the first part using a linear regression model. Estimates for equation 1 are reported in Table 2. We start with the specification that excludes all state-level controls. Column 1 indicates that a $100 increase in weekly UI benefits reduces the likelihood of being food insecure by 1.9 percentage points. On the basis of the 41% of UI-eligible households that have experienced some food difficulty, this

translates to a reduction of approximately 4.6% (calculated as the coefficient divided by the sample average of the dependent variable, $\frac{0.019}{0.410}$).[15]

The controlled state fixed effects can account for the time-invariant factors across states that may correlate with unemployment benefits and affect food insecurity. However, if some states experience economic downturns, they are likely to have both high UI benefits and large numbers of food-insecure households. To address this concern, column 2 of Table 2 includes a series of state-level economic attributes to control for local economic conditions and demographic compositions. We find a slight decrease in the estimated maximum weekly UI benefits to -0.016. In column 3, where we further control for a set of state welfare policies and programs, the significant reduction of 2.5 percentage points is consistent with those of the first two columns, and further adding state-specific linear trends yields a statistically significant estimate of -0.023 in column 4. If we use the estimates in the last column, a $100 increase in maximum weekly UI benefits reduces the probability of being food insecure by approximately 2.3 percentage points or approximately 5.6%.

We further estimate the UI effects on the level of food insecurity among households reporting at least one difficult condition regarding food intake by using the second part of the model, specified as follows:

$$(2)\quad \mathrm{E}(s_{ist}|y_{ist} > 0) = \alpha_0 + \alpha_1 B_{st} + \alpha_2 R_{st} + X_{ist}\eta_1 + Z_{st}\eta_2 + (\eta_s t) + \tau_t + \delta_s$$

The covariates are the same as those in equation 1, but we now restrict the sample to households whose food insecurity status is coded as 1 (i.e., $y_{ist} > 0$). The outcome variable in the second part, denoted by s_{ist}, is the Rasch food security scores provided in the survey. Similar to the first part

estimated in Table 2, we augment the regression function with more state-level controls from column 1 to column 4 in Table 3. We consistently identify negative effects of increasing maximum weekly UI benefits on the severity of food insecurity for UI-eligible households who report having experienced some food insufficiency. Specifically, the coefficient in column 4 that is at the upper bound of the estimated effects suggests that a $100 increase in maximum weekly UI benefits reduces the conditional severity of food insufficiency by approximately 6.3%.

Our sample period includes the Great Recession, during which emergency expansions occurred in both the levels of UI benefits and durations for claiming these benefits; adjustments in many other public assistance programs also likely occurred. To determine whether our results are driven by the inclusion of this unusual period, we re-estimate the two-part model excluding the Great Recession period (2008-2009) in Table 4. We confirm that the negative effects on both the likelihood and the conditional severity of household food insecurity are insensitive to the exclusion of the Great Recession period.

SNAP Enrollment and Claimed Benefits

We now analyze the effects of a more generous UI on household decisions to enroll in SNAP and the benefits received. In this subsection, we limit our sample to households with members who are UI recipients. As in the previous subsection, we exploit the variation in maximum weekly UI benefits across states and over time and adopt the two-part model specified in equations 1 and 2 to account for approximately 86% of UI recipients in our sample who are not SNAP beneficiaries and thus receive zero SNAP benefits. In the first part of the model, we examine the UI effect on the likelihood of SNAP enrollment, where the outcome variable is coded as 1 if one or more

household members have received SNAP benefits over the past 12 months and 0 if no one in the household has claimed SNAP benefits.[16] In the second part, we estimate the UI effect on the value of SNAP benefits received by participating families.[17] We use the March CPS weights for both parts of the model.

The results of the first and second parts are reported in Tables 5 and 6, respectively. We add state economic conditions and welfare policies sequentially to the regressions, and specifications including such state-level controls typically find smaller effects of an UI expansion. Overall, we consistently identify evidence of program substitution between UI and SNAP. In particular, a $100 increase in maximum weekly UI benefits reduces the probability of enrolling in SNAP by approximately one percentage point, equating to approximately 7.1%. Among the households benefiting from both UI and SNAP, a $100 increase in maximum weekly UI benefits reduces SNAP benefits by about $178 per year, which is equal to a reduction of 6.2%.

Event-study Design

In both parts of the two-part model, we adopt the conventional difference-in-differences (DD) technique, in which we compare changes in outcomes between households in states where maximum weekly unemployment benefits expanded and households in states where maximum weekly UI benefits did not change. This model may not be able to fully control for the heterogeneity in state-level characteristics; therefore, the parallel trend assumption could be violated. The inclusion of state-specific linear time trends might be helpful in addressing this concern to some extent. Furthermore, we adopt an event-study design to further examine whether we have evidence that the parallel trend assumption is violated. In accordance with Cotti, Nesson,

and Tefft (2018), we conduct a pseudo-event study that accommodates a continuous treatment varying in magnitude (i.e., increases in maximum weekly UI benefits in our case). As demonstrated in the related literature (e.g., Mueller, Rothstein, and Von Wachter 2016), the precision of these estimates might be affected by repeated state-level UI expansions.

Recognizing these limitations, we estimate a dynamic regression with three leads and three lags of the treatment variable (B_{st}) that spans 7 years around a UI expansion. The regression is specified as follows:

$$(3)\quad Outcome = \beta_0 + \sum_{k=-3}^{3} \alpha_k B_{s,t-k} + \beta_1 R_{st} + X_{ist}\eta_1 + Z_{st}\eta_2 + (\eta_s t) + \tau_t + \delta_s + \varepsilon_{ist}$$

The covariates are the same as those in equations 1 and 2. We estimate both parts of the model using this equation, and results are reported with and without the state-specific linear trends ($\eta_s t$). Figure 1 plots the coefficients of maximum weekly UI benefits on both the extensive and intensive margins of household food insecurity and the corresponding 90% confidence intervals for the 3 years leading up to and 3 years after a UI expansion. Similarly, Figure 2 presents the coefficients and corresponding 90% confidence intervals for the likelihood of SNAP enrollment and claimed SNAP benefits.

There are two specific goals underlying this analysis. First, the leading coefficients allow us to detect the pre-existing trends in the outcomes and assess the validity of the DD design. We would expect them to be undistinguishable from zero if households surveyed prior to an UI expansion do not react to the policy change that has not been taken place. It is similar to a placebo test on the alternative periods when the UI has not expanded. However, if the UI expansion is correlated with some pre-existing state-level unobserved characteristics that affect household food insecurity or

SNAP usages, the leading variables would be statistically significant. Second, it also helps examining the lagged effects of UI expansions.

Two results stand out from Figures 1 and 2. First, no economically meaningful pre-trend emerged among any of the outcome variables. The probability of being food insecure among the UI-eligible households tends to slightly decrease prior to UI expansions, but this downward trend disappears when state-specific linear time trends are included. We also identify slight decline in the other outcome variables immediately before a UI expansion, but the decline is either corrected by the inclusion of state trends or is economically and statistically negligible. We further confirm the joint insignificance of all leading variables.[18] Overall, we find no concerning trends prior to UI expansions.

Second, an immediate reduction in all four outcome variables occurs when state maximum weekly UI benefits increase. However, neither the improvement in household food intake nor the reduction in SNAP usages persist over time. We also confirm the joint insignificance of the post-expansion effects (excluding the year of the expansion).[19] These effects are not long-lasting possibly because the UI benefits do not generate lasting effects on household financial status. For one thing, the benefit by nature is transitory. The explicit objective of the UI system is to provide temporary income assistance for unemployed workers and most states had a maximum duration of 26 weeks for receiving unemployment benefits. For another thing, although the empirical evidence is mixed, some studies find that UI expansion does not appear to significantly improve one's labor market outcomes, such as re-employment wages. Nazarov (2009) finds that re-employment wages decline with increases in the UI wage replacement rates; McCall and Chi (2008) look into the dynamics and find that increases in the UI benefits have positive effects on the re-employment

wages in the short run, but then dissipate with the length of the unemployment spell, and even become negative as unemployment progresses further.

Robustness Analyses

The primary results reveal evidence of two social benefits of more generous UI: improved household food intake among UI-eligible households and cost savings on SNAP in terms of both the number of enrollees and benefits claimed. In this subsection, we conduct a series of robustness checks using alternative specifications and different empirical approaches.

Triple-differences model

To account for any unobserved state-level factors that were omitted from our primary regressions and potentially affect the outcome variables, we adopt a within-state comparison group by using a triple-differences (DDD) model. Following the common practice in the literature (e.g., Fu and Liu 2019; Kuka 2020; Swensen, Lindo, and Regmi 2020), we use UI-ineligible households, defined as families with no unemployed members, to be the within-state control group based on the assumptions that these families are not affected by changes in unemployment benefits and that their trends in the outcome variables are parallel with those of the primary samples. Hsu, Matsa, and Melzer (2018) also use the employed as the counterfactuals in studying the impacts of UI on housing market.[20]

Specifically, for the analyses of household food insecurity, we add an indicator that is coded as 1 if a household is eligible for UI and 0 if it has no unemployed members and is thus UI-ineligible. For the estimates of SNAP usage, we code the indicator as 1 for UI recipients and 0 for UI-ineligible households. We further include all pairwise interactions. The results are reported in

Tables 7 and 8. We confirm that the negative effects on both margins of household food insufficiency and SNAP enrollment are virtually identical with the DD estimates, and the impact on the value of SNAP benefits are slightly larger in magnitude.

Alternative samples

Our baseline sample includes all households with at least one member who reports being unemployed some time during the past 26 weeks and are not re-/new entrants. We test whether our baseline results are sensitive to a series of alternative sample choices. First, we extend the restrictions on unemployment spells from 26 weeks to 52 weeks; the food insecurity questions are based on this longer period. Second, we exclude unemployed people who are job leavers as they might be voluntarily unemployed and hence ineligible for UI benefits. We further remove temporary layoffs and retain only those who involuntarily lost their jobs because they are most likely to meet the requirement of becoming unemployed by "no fault of one's own". We finally include all households regardless of their reasons for unemployment.

The results in the online supplementary appendix Table A1 consistently indicate strong evidence of reductions in both food insufficiency and SNAP usage associated with more generous UI, and the results are robust to the various choices of sample. In particular, the effects are slightly smaller when we include all types of unemployed households in columns 11-14, and the results do not differ considerably from our baseline estimates when we focus on the sample of people who involuntarily lost their jobs in columns 7–10.

In the primary analyses of program interactions, we include all UI recipients with positive unemployment benefits. As a sensitivity test, we exclude households whose claimed UI benefits are less than $100 per year to account for any reporting errors. We confirm that the baseline estimates are insensitive to the alternative choice of sample when we add the $100 requirement in

Tables A2 and A3. Both the effects on SNAP claims and the reduction in conditional SNAP benefits are virtually the same as the baseline results.

Balance test

One concern regarding the main analyses is that the estimates may suffer from sample selection bias. Because we include only UI-eligible families in the analysis of food insecurity, if the composition of the unemployed population is influenced by changes in the UI policy, the sample would be stratified on an endogenous variable. Similarly, because we restrict the sample to actual UI recipients in the analysis of SNAP usage, if UI expansion induces a change in the composition of UI enrollees, our estimates would be biased.

To address this concern, we conduct a series of balance tests, the results of which are shown in Table A4. We test whether the demographic characteristics of the studied households are affected by UI policy changes. Specifically, we regress a series of demographics on state maximum weekly UI benefits along with other controls as in our baseline model respectively for the analysis of food insecurity using the UI-eligible sample and the analysis of SNAP utilization based on the UI-recipient sample.

Overall, we do not observe a large effect of UI expansion on the composition of either sample. Estimates of UI benefits on family size among UI recipients are sometimes statistically significant. However, the results are sensitive to the inclusion of state-specific linear trends. In addition, the age of the UI-eligible head of household appears to decline as UI expands. A $100 increase in maximum weekly UI benefits is associated with a decrease in the age of UI-eligible head of household by approximately 0.47–0.61 years. Although the effect is statistically significant, we interpret this effect to have little economic significance because of its small magnitude. Because

most demographic characteristics appear to be unaffected by UI expansion, we do not make this a major concern in our study.[21]

Alternative measures of UI benefits

The US Department of Labor releases the Comparison of State Unemployment Insurance Laws in January and July of each year. Throughout the analyses, we use the maximum weekly UI benefits reported in July and match them with the dependent variables by year. However, because food security status and SNAP usage are surveyed at a 12-month interval, households might respond to UI policy changes in both January and July. Therefore, as a robustness test, we use the 2-month averages of the maximum weekly UI benefits reported in a specific year and evaluate whether our estimates are sensitive to different measures of UI benefits. The results of household food insecurity are reported in Table A5, and estimates of SNAP utilization are presented in Table A6. The coefficients are broadly consistent with the baseline results, suggesting that the estimates are robust to this alternative measure.

Furthermore, in the first part of our model, we define a household as food insecure if it reports some difficulty in food intake (i.e., at least 1 affirmative answer to the 18 related questions). However, the binary measure typically used in the literature considers households as being food insecure only when they have at least three affirmative answers in the survey. Therefore, we may overstate household food insecurity by including those who answer affirmatively to only one or two questions. As a robustness check, we adopt the binary measure commonly used in the literature for comparison with our first-part estimates.

Results in Table A7 suggest that a $100 increase in maximum weekly UI benefits, which raises household annual UI income by about $460 (see Table 9), is associated with a decrease in the likelihood of household food insecurity by about 10.4%. Schmidt, Shore-Sheppard, and Watson

(2016), however, show that raising a family's imputed potential food and cash benefits by $1,000 reduces their food insecurity by about 3.3%. Our results are somewhat larger than theirs. The disparity could arise from (1) different choices of sample: Schmidt, Shore-Sheppard, and Watson (2016) study the low-income single-parent families during the 2001-2009 period, while we focus on UI-eligible households over the 1999-2017 period; and (2) different effects identified: the 2SLS strategy adopted in Schmidt, Shore-Sheppard, and Watson (2016) identifies a Local Average Treatment Effect for the compliers whose benefits are affected by changes in the state policy parameters, while our estimates identify the Intention-to-treat (ITT).

Finally, all regression models so far are linear in levels. We further test whether our estimates are robust to different functional forms by adopting a simple level-log model. Specifically, we use the logarithm of state maximum UI benefits. The estimates for household food insecurity are reported in Table A8, and the interaction with SNAP is presented in Table A9. The level-log results are qualitatively similar to our primary findings that use the linear model, and we continue to observe significant reductions in both the margins of household food insecurity among the UI-eligible families and evidence of program substitution with SNAP when UI benefits increase.

Subsample analyses

To check whether the main results hold for different subsamples, we first consider the UI expansion impacts separately for households with and without small children under age five. Results on household food insecurity and SNAP usages are reported respectively in the top two panels of Tables A10 and A11. We find that the reduction in household food insecurity associated with UI expansion hold for both groups, while the UI impacts on household SNAP utilization appear to be slightly stronger for those with small children. We also separate our estimation based on household education levels. In particular, we divide the sample into two groups: households

with members who have above-high-school education and those with no members who have at least high school education. Results in the bottom two panels of Tables A10 and A11 show that the relatively less-educated families tend to benefit more from the UI expansion. Specifically, we find reduction in both margins of food insecurity and decline in the SNAP value associated with UI expansions for the less-educated households, while the UI impacts are substantially smaller for the more-educated counterparts in terms of both the magnitudes and the statistical significance.

Placebo tests

Finally, we conduct a series of placebo tests to further check whether our baseline results are driven by some unobserved state-level factors omitted from our primary regressions. Specifically, we run our baseline regressions based on the UI-ineligible households. Since they are not qualified for UI benefits, we should expect null effects of UI expansion on household food insecurity among them. In particular, we restrict our sample to households whose unemployed members are only consisted of the re-/new entrants, and report the results in Panel A of Table A12. We limit to households whose unemployed members are only consisted of job leavers with unemployment durations exceeding 26 weeks in Panel B, and combine the two types of households in Panel C. We also restrict our sample to households that have members who are out of the labor force but no UI-eligible members in Panel D. Overall, we find null effects among the UI-ineligible households, which confirm the robustness of our baseline results.

Discussion

Mechanism

Our primary analyses reveal that an increase in UI benefits leads to a reduction in both household food insecurity and SNAP utilization. These results are likely driven by a positive income effect by which the cash benefits of UI expand a household's budget set, thereby improving food intake among UI-eligible families and reducing the SNAP benefits received by UI enrollees. To assess this potential channel, we estimate how household's claimed value of UI is affected by changes in the maximum weekly UI benefits. The estimates in Table 9 confirm the positive income effect. Specifically, a $100 increase in maximum weekly UI benefits increases household annual UI income by about $460, a rise of approximately 7.2%.

Aside from the two-part model that separately examines the extensive and intensive margins of household food insecurity, we further explore alternative classifications and study whether the improvement in food insufficiency associated with UI expansions is mainly driven by households in severe food shortages or those who are moderately food insecure. We utilize the food security scale reported in the CPS-FSS that classifies households as being low food secure if they report 3-7 food insecure conditions, very low food secure if they report 8 or more food insecure conditions, and food secure if otherwise. We define three binary variables describing the movement from food security to low food security, from food security to very low food security, and from low food security to very low food security, respectively. We confirm that such classification divides our sample in a compatible manner as in Ratcliffe, McKernan, and Zhang (2011).

Results in Table A13 suggest that the impacts are mostly driven by the severely food insecure households who move away from being very low food secure to little or no food shortages associated with UI expansions, while those who are moderately food insecure do not appear to respond significantly to the increases in UI benefits. To be specific, a $100 increase in maximum weekly UI benefits substantially reduces the probability of transiting from food security to being

severely food insecure by approximately 25%, and the likelihood of transiting from low food security to being severely food insecure by about 13.6%. Our findings are not uncommon in the literature. For example, Schanzenbach et al. (2016) find similar heterogeneous impacts of the Summer Electronic Benefits Transfer for Children (EBT) on household food insecurity. They show that a $60 per month Summer caused a more than 30 percent reduction in very low food security among households with children, while the reduction in moderate food insecurity was less than 20 percent.

The disparate results are likely driven by nonlinear income effects that are larger for those who are in greatest needs and smaller for households with less severe food shortages. In fact, households with severe food insufficiency are disproportionally low-income. According to the USDA, about 15% of the low-income households suffer from severe food insecurity, while that share among all households is about 5%.[22] Empirical evidence has uncovered that cash transfer programs tend to have larger impacts among the lowest-income households, such as the Mothers' Pension studied in Aizer et al. (2016), Moreover, unlike the SNAP benefits, households may not necessarily spend their UI income on food consumption, especially those who are higher-income and suffer less from food insufficiency. Studies show that the cash transfers to the relatively higher-income households appear to have stronger effects on the diet diversity and much smaller effects on the quantity of food consumed (e.g., Cunha 2014; Hidrobo et al. 2014; Hoddinott, Sandström, and Upton 2018). Therefore, it may not be surprising that the UI expansion benefits households who are in severe food insecurity more than their counterparts who suffer less from food insecurity.

Welfare Analysis

In this subsection, we assess the economic value of substitution between UI and SNAP by conducting a welfare analysis. Specifically, we calculate the expenditures saved on SNAP as a result of a more generous UI by using the estimates obtained in the previous sections.

An increase in maximum weekly UI benefits changes both the extensive and intensive margins of SNAP utilization. First, on the extensive margin, some households become ineligible for SNAP because of increased family income. By combining the numbers of US households published annually by the Census Bureau with the share of households receiving UI in the March CPS, we calculate that an average of approximately 6,684,513 households claim UI benefits every year during our sample period, among whom about 13.7% are also enrolled in SNAP, according to Table 1.[23] Estimates from Table 5 suggest that a $100 increase in maximum weekly UI benefits reduces the likelihood of enrollment in SNAP by approximately one percentage point from 13.7% to 12.7%. As a result, about 66,845 households will stop claiming SNAP benefits. Those who remain enrolled in SNAP receive an average annual benefit of $2783.568, as reported in Table 1. Therefore, on the extensive margin, a $100 increase in maximum weekly UI benefits results in SNAP savings of approximately $186 million (=$2783.568×66,845).

Second, UI expansion affects SNAP usage on the intensive margin because households who remain enrolled receive fewer benefits. The estimates in column 4 of Table 6 suggest that a $100 increase in maximum weekly UI benefits reduces SNAP benefits by $177.983. Moreover, the 12.7% of UI recipients who remain enrolled in SNAP after UI expansion each year translates to approximately 848,933 households (=6,684,513×0.127). Thus, on the intensive margin, SNAP expenditure is reduced by about $151 million (=$177.983×848,933). The sum of savings on both margins results in a reduction in total SNAP expenditure by about $337 million. It is noteworthy

that as shown in the event study design, such benefits are mainly from the year when UI expands as opposed to any lagged effects.

Subsequently, we weigh the calculated savings against the increase in UI costs associated with a $100 increase in maximum weekly UI benefits. To evaluate the effects of such UI expansions on its total costs, we analyze the information provided by the US Department of Labor's Comparison of State Unemployment Laws regarding state-level per-person weekly UI expenses and annual UI expenditures. We then run regressions at the state level with these expenses as the outcome variables while controlling for state welfare programs and characteristics along with state and year fixed effects.

As presented in Table 10, an increase in maximum weekly UI payments by $100 leads to increases of about $23.732 in state per-person weekly UI expenditures and around $164 million in state yearly total UI expenses. After combining the former estimator with the annual average of the number of weeks compensated for all UI recipients, we find an increase of approximately $3,259 million in total annual UI expenditures associated with UI expansion.[24] The latter estimate on state annual UI expenditures suggests a much larger increase of approximately $8,364 million nationwide (=$164 million×51). The discrepancy between the two estimates is likely caused by the potential effect of UI expansions on the number of UI participants, which the estimates on state per-person weekly UI expenditures ignore. If the increase in unemployment benefits expands the pool of UI recipients, then the per-person expenditure result that does not account for such an effect would understate the direct costs of UI expansion, resulting in a lower bound of the relative cost savings on SNAP.[25] Therefore, for every dollar spent on UI, the savings on SNAP range from $0.04 (s.e. 0.017) to $0.11 (s.e. 0.038).[26] The parallel analysis by Lindner (2016) shows that about

15 cents can be saved on disability insurance expenses for every dollar increase in UI spending. Our results reveal a smaller substitution effect with SNAP.

Conclusion

The literature has widely documented the consumption-smoothing benefits of social insurance programs. Although these programs, such as UI, mitigate the reductions in food consumption upon unemployment, they do not necessarily improve food intake, which motivated our paper that quantifies the effect of social insurance on household food hardship. In particular, we show that a more generous UI significantly reduces food insecurity among UI-eligible households. We then included the information on household income and provide evidence of a positive income effect through which UI expansion likely influences household food insufficiency.

The economic value of UI expansion is not yet settled. We emphasize the role of interaction between UI and related programs, especially those designed to reduce food insecurity. This paper studies the interaction of UI with a major food assistance program, namely SNAP, and presents strong evidence of program substitution. Specifically, an increase in UI benefits reduces SNAP utilization on both the extensive and intensive margins. This result is valuable because the spending saved on SNAP as a result of more generous UI reinforces the social benefits of UI expansion. If, by contrast, the estimates suggested that the programs are complementary, the benefits of reducing food insecurity would be compromised because of the increased costs of SNAP. Program interactions are crucial for calculating the optimal UI provision level, as emphasized by Lawson (2015), and they can be useful for assessing social programs in general.

Overall, our results indicate that the generosity of UI reduces household food insecurity and saves spending on SNAP. Although beyond the scope of this paper, the incorporation of these results into UI calculation to analyze how they change the optimal level of UI benefits is a promising subject for future work.

Endnotes

[1] Similar effects are also found in other social programs including the workers' compensation for the injured labor (Bronchetti 2012), social security benefits for disabled individuals (Meyer and Mok 2019), and social insurance and transfers for individuals subject to health shocks in the Swedish context (Kolsrud, Landais, and Spinnewijn 2019).

[2] We note that the two-part model does not make any assumption about the correlation between the errors of the binary equation in the first part and the continuous regression in the second part (Belotti et al. 2015). Therefore, the estimates in the second part should be interpreted as conditional impacts based on the sample of households with positive values in the level of food insecurity, rather than unconditional impacts for all.

[3] Both the two-part model and the sample selection model (Heckman 1979) are used to address outcome values clustering at zero. The selection model is often used in wage equation estimation. In such applications, we don't observe one's wage if the individual doesn't work; and thus, the selection model estimates both actual and potential workers. However, there is unlikely to be a latent positive expected level of being food insecure. Therefore, what we are trying to model is actual food insecurity, as opposed to potential food insecurity.

[4] See the Comparison of State Unemployment Laws reports provided by the U.S. Department of Labor https://oui.doleta.gov/unemploy/pdf/uilawcompar/2019/complete.pdf.

[5] The baseline sample consists of both job losers and job leavers. Although job losers are the dominant group in the UI-recipient population, about 18.7% of the job leavers also apply for the UI benefits and around half of them receive positive UI benefits (Vroman 2009). Therefore, to consider a broad sample of households who are potentially affected by UI expansions, we choose to keep the job leavers in the sample.

[6] Three questions are asked to household as a whole and seven are asked to the adults in the households. If there is a child present in the family, there will be eight additional questions. Therefore, households with no children are asked with 10 questions and households with children are asked with 18 questions in total. The details can be found in Table S-5 in the Statistical Supplement of Coleman-Jensen et al. (2019).

[7] See Andrich (1988), Moffitt and Ribar (2016), and Bickel et al. (2000) for more on Rasch model.

[8] See Appendix Table 10 in Gundersen (2008).

[9] This is equivalent to classifying marginally food-secure households as being food insecure. We later provide a sensitivity test by adopting the commonly-used binary measure for food insecurity status, and compare with our first-part results.

[10] We recognize the possibility that, in the way we define the sample, we may throw out some households who are actually receiving the UI but underreport their enrollment status. Unless increases in the UI parameters cause this underreporting error to shift or change the composition of families who underreport their UI recipiency, it would be less of a concern in this paper.

[11] Specifically, the topcodes in the UI benefit variable are 9,999 before 2008 and 24,999 afterwards in the March CPS.

[12] http://workforcesecurity.doleta.gov/unemploy/statelaws.asp#Statelaw. Throughout the analyses, we use the maximum weekly benefit amounts including dependent allowances.

[13] This variable is classified into three categories: call centers available statewide; call centers available only in selective parts of the state; and no call centers, which is considered as the reference group.

[14] This variable is classified into three categories: asset limit is increased but not eliminated; asset limit is eliminated; and no broad-based categorical eligibility, which is considered as the reference group.

[15] Throughout the paper, we do not report the coefficients on maximum UI duration simply because there is little variation in maximum UI durations across states over our sample period between 1999 and 2017 except for the Great Recession period, and hence it has virtually no impacts on household food insufficiency and SNAP usages.

[16] Survey questions regarding the receipt of means-tested benefits, including SNAP, are only asked to households whose income is below a certain threshold. Households with income above the threshold are considered to be not qualified for the programs and thus coded as 0 for not participating in the SNAP.

[17] The dependent variable may suffer from measurement error as the CPS is often criticized by understating the number of SNAP enrollees and the value of benefits received (e.g., Tiehen, Jolliffe, and Smeeding 2015). We note that this would be less of a concern if the measurement error is arguably uncorrelated with UI expansions or remains constant over time. Reich and West (2015) argue that the misreporting rates follow a linear trend, which will then be absorbed by the inclusion of state-specific linear time trends.

[18] To be specific, in Figure 1, The tests on the joint significance of all leading variables for food insecurity status return p-values of 0.274 and 0.464 with and without state-specific time trends, and respectively 0.109 and 0.108 for the conditional severity of food shortages. In Figure 2, we obtain p-values of 0.416 and 0.290 with and without state-specific trends, and respectively 0.388 and 0.379 in estimating the value of SNAP benefits.

[19] In Figure 1, the tests on the joint significance of all lagged variables for food insecurity status return p-values of 0.978 and 0.986 with and without state-specific time trends, and respectively 0.308 and 0.977 for the conditional severity of food shortages. In Figure 2, we obtain p-values for all lag variables in estimating SNAP enrollment at 0.274 and 0.131 with and without state-specific trends, and respectively 0.987 and 0.936 in estimating SNAP value.

[20] In concern that the employed might not be comparable with the unemployed households, we examine the DDD model using alternative within-state control groups who are also unemployed but potentially ineligible for receiving UI benefits. Specifically, we use the households whose unemployed members are only consisted of the re-/new entrants, households whose unemployed members are only consisted of job leavers with unemployment durations exceeding 26 weeks, and the two types of households combined. We confirm that the DDD results are similar with those reported in Table 7.

[21] To further address the concern regarding self-selection into unemployment, we also examined whether UI expansion affects unemployment duration and the likelihood of households being unemployed. For unemployment durations, we look into both *total* and *average* unemployment weeks at the household level. For the incidence of unemployment, we first use our baseline sample and study whether UI expansion would lead more family members

to unemployment within the UI-eligible households. We then include all families and examine whether households select into the UI-eligible sample associated with higher UI benefits. Overall, we do not find any significant results for all the above outcomes.

[22] https://www.ers.usda.gov/data-products/chart-gallery/gallery/chart-detail/?chartId=58384.

[23] All statistics in Table 1 are weighted based on the corresponding sampling weights.

[24] We use the quarterly statistics on the total weeks compensated provided by the U.S. Department of Labor. This measure adds up the total number of weeks compensated for all UI recipients, during which they claim the UI benefits in each quarter. After converting to an annual average of 137,327,469, we obtain the change in the total UI expenses associated with a $100 rise in the maximum weekly UI at $3,259 million (=$23.732×137,327,469).

[25] There is evidence that expansions in UI benefits significantly increase the UI takeup rates (e.g., McCall 1995; Anderson and Meyer 1997).

[26] The lower and upper bound estimates are nonlinear combinations of the point estimates of the UI expansion effects on SNAP usage (column 4 in Table 5 and column 4 in Table 6), and on state UI expenditures (columns 2 and 4 in Table 10). Standard errors are obtained by simultaneously estimating the above models and then using the delta method to test nonlinear hypotheses about estimated coefficients.

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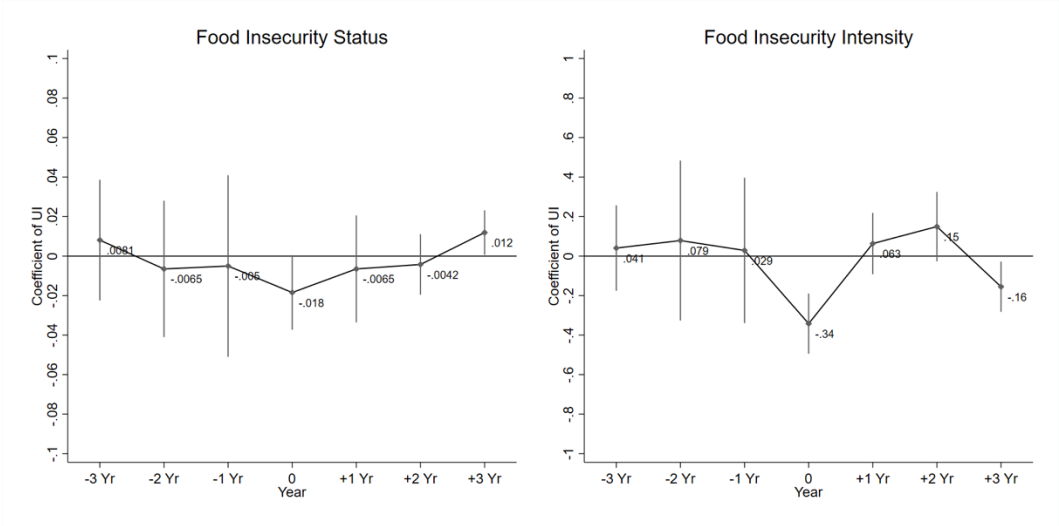
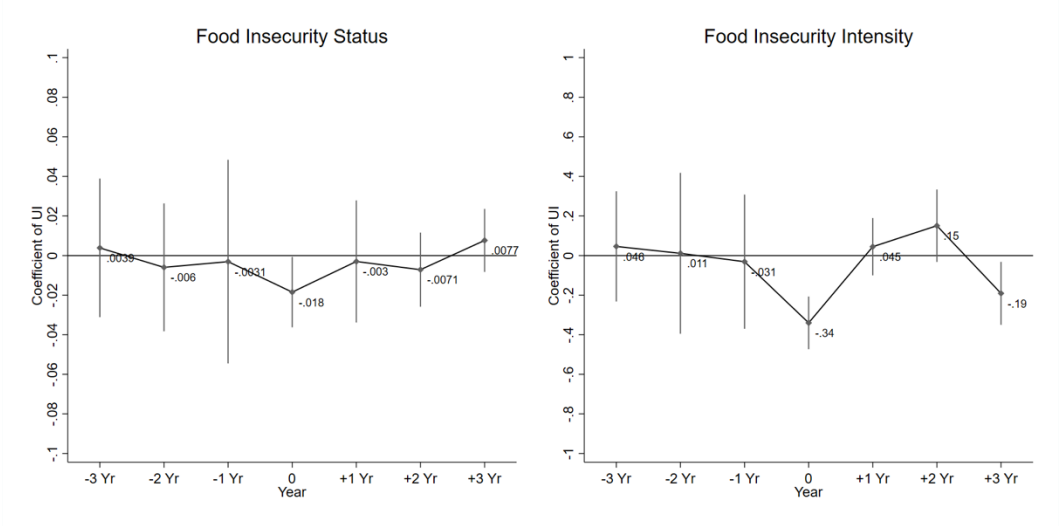


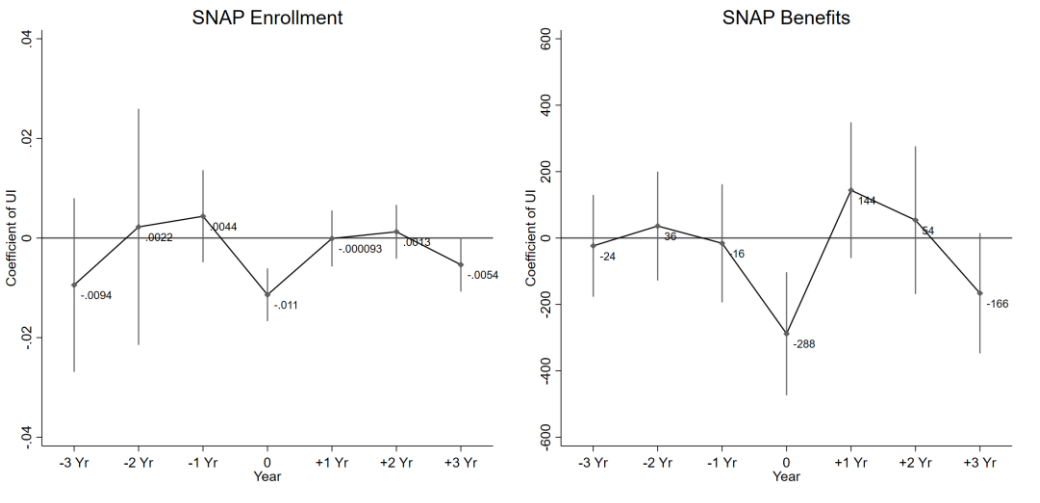
A. Without state-specific linear trends



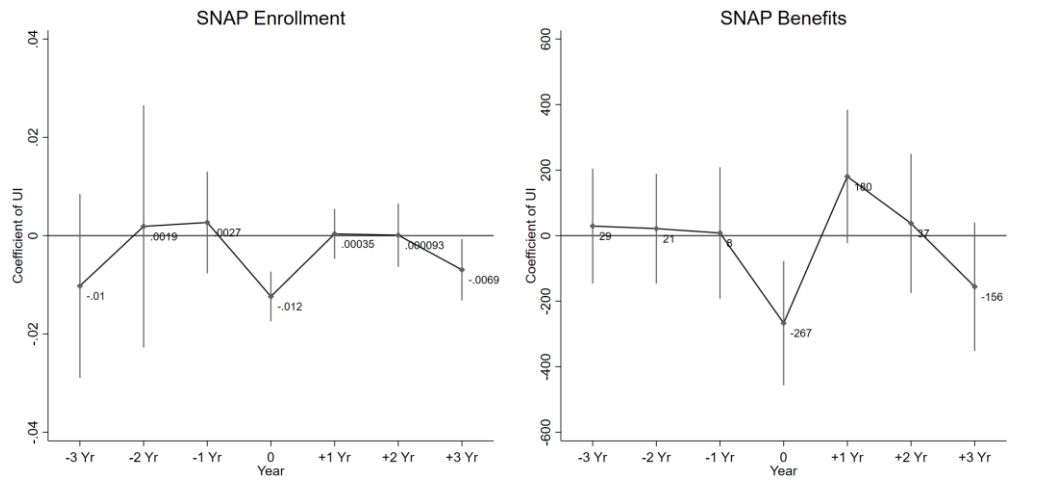
B. With state-specific linear trends

Figure 1. Pseudo-event study for the UI effect on household food insecurity

Note: This figure uses a pseudo-event study in equation 3 with the CPS-FSS weights. Panel A reports the estimates without state-specific linear time trends and Panel B uses estimates with the inclusion of state trends. The markers show the point estimates by event date, with the value of coefficients written by the side. The error bars indicate the 90% confidence intervals around the point estimates, with the standard errors clustered at the state level. The tests on the joint significance of all leading variables for food insecurity status return p-values of 0.274 and 0.464 with and without state-specific time trends, and respectively 0.109 and 0.108 for the conditional severity of food shortages. The tests on the joint significance of all lagged variables for food insecurity status return p-values of 0.978 and 0.986 with and without state-specific time trends, and respectively 0.308 and 0.977 for the conditional severity of food shortages.



A. Without state-specific linear trends



B. With state-specific linear trends

Figure 2. Pseudo-event study for the UI effect on SNAP usage

Note: This figure uses a pseudo-event study in equation 3 with the March CPS weights. Panel A reports the estimates without state-specific linear time trends and Panel B comes from regressions controlling for state trends. The markers show the point estimates by event date, with the value of coefficients written by the side. The error bars indicate the 90% confidence intervals around the point estimates, with the standard errors clustered at the state level. The tests on the joint significance of all leading variables in estimating SNAP enrollment return p-values at 0.416 and 0.290 with and without state-specific trends, and respectively 0.388 and 0.379 in estimating the value of SNAP benefits. We obtain p-values for all lagged variables in estimating SNAP enrollment at 0.274 and 0.131 with and without state-specific trends, and respectively 0.987 and 0.936 in estimating SNAP value.

Table 1. Summary Statistics

	(1)	(2)	(3)	(4)
	CPS-FSS		March CPS	
Variable	Mean	Std Dev.	Mean	Std Dev.
Household food insecurity status	0.421	(0.494)		
Household food insecurity level[a]	4.773	(2.550)		
Household food insecurity status (alternative)[b]	0.280	(0.449)		
Household SNAP take-up status			0.137	(0.344)
Household SNAP market value[c]			2783.568	(2314.719)
Household actual UI income			3964.154	(6131.062)
Household head age	42.663	(13.365)	43.743	(12.546)
Household share of white members	0.756	(0.416)	0.792	(0.393)
Marital status	0.493	(0.500)	0.545	(0.498)
Share of primary-aged members with high school education	0.301	(0.325)	0.279	(0.327)
Share of primary-aged members with some college education and above	0.308	(0.348)	0.338	(0.354)
Share of members aged 20-60	0.698	(0.269)	0.707	(0.271)
Single-person household (N=1)	0.114	(0.317)	0.142	(0.350)
Regular-sized household (N=2-5)	0.801	(0.399)	0.798	(0.401)
Presence of child under age 6	0.244	(0.429)	0.233	(0.423)
Observations	27,670		78,991	

Note: The first two columns are based on the CPS-FSS data and restrict to the sample of UI-eligible households. To arrive at this sample, we make two restrictions. First, we restrict our baseline sample to families with at least one member who was unemployed at any time during the past 26 weeks. Second, we exclude the re-/new entrants because they are unlikely to have positive earnings in base periods. Weighted summary statistics for the outcome and household demographic variables are reported using the CPS-FSS weights. The last two columns are based on the March CPS data and we restrict the sample to actual UI recipients, defined as those with positive unemployment benefits. We exclude individuals who report a topcoded value of unemployment benefits. Weighted summary statistics for the outcome and household demographic variables are reported using the March CPS weights. Throughout the analyses, we exclude households with no members aged 20-60 years. All monetary variables are adjusted by the Food CPI to 2018 dollars.

[a] Household food insecurity level is only measured for those who answers affirmatively to at least one of the 18 food insecurity questions in the CPS-FSS.

[b] This measure follows the literature by assigning 0 to the households who have high or marginal food security, and 1 to the households who have low or very low food security.

[c] Household SNAP market value is only measured for the households who have enrolled in SNAP.

Table 2. Effect of Unemployment Insurance on Food Insecurity Status, 1st Part

	(1)	(2)	(3)	(4)
Variable	Food insecurity status			
Maximum weekly UI benefit (in $100)	-0.019**	-0.016*	-0.025***	-0.023**
	(0.007)	(0.009)	(0.009)	(0.010)
Observations	27,670	27,670	27,670	27,670
Mean dependent variable	0.410	0.410	0.410	0.410
State-level welfare policies			Y	Y
State-level economic attributes		Y	Y	Y
Household characteristics	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
State-specific linear trend				Y

Note: This table estimates the impact of an increase in the maximum weekly UI benefits on household food insecurity status using equation 1 with the CPS-FSS weights. Results on maximum UI durations, household characteristics and state-level controls are not reported. Standard errors are in parentheses and clustered at the state level. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 3. Effect of Unemployment Insurance on Food Insecurity Level, 2nd Part

	(1)	(2)	(3)	(4)
Variable	Food insecurity level			
Maximum weekly UI benefit (in $100)	-0.151***	-0.192***	-0.204***	-0.301***
	(0.042)	(0.044)	(0.053)	(0.067)
Observations	11,343	11,343	11,343	11,343
Mean dependent variable	4.754	4.754	4.754	4.754
State-level policies			Y	Y
State-level economic attributes		Y	Y	Y
Household characteristics	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
State-specific linear trend				Y

Note: This table estimates the impact of an increase in the maximum weekly UI benefits on the level of household food insecurity using equation 2 with the CPS-FSS weights. We restrict the sample to households who report some experience of food shortage. Results on maximum UI durations, household characteristics and state-level controls are not reported. Standard errors are in parentheses and clustered at the state level. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 4. Effect of Unemployment Insurance on Food Insecurity: Removing Great Recession

	(1)	(2)	(3)	(4)
Variable	Food insecurity status		Food insecurity level	
Maximum weekly UI benefit (in $100)	-0.026**	-0.023**	-0.185***	-0.296***
	(0.011)	(0.011)	(0.067)	(0.084)
Observations	23,534	23,534	9,517	9,517
Mean dependent variable	0.404	0.404	4.719	4.719
State-level policies	Y	Y	Y	Y
State-level economic attributes	Y	Y	Y	Y
Household characteristics	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
State-specific linear trend		Y		Y

Note: This table estimates the impact of an increase in the maximum weekly UI benefits on both the likelihood and the conditional severity of household food insecurity using equations 1 and 2 with the CPS-FSS weights, excluding the Great Recession period. Results on maximum UI durations, household characteristics and state-level controls are not reported. Standard errors are in parentheses and clustered at the state level. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 5. Effect of Unemployment Insurance on SNAP Enrollment, 1st Part

	(1)	(2)	(3)	(4)
Variable	SNAP enrollment			
Maximum weekly UI benefit (in $100)	-0.019***	-0.017***	-0.011***	-0.010**
	(0.006)	(0.005)	(0.003)	(0.004)
Observations	78,991	78,991	78,991	78,991
Mean dependent variable	0.141	0.141	0.141	0.141
State-level policies			Y	Y
State-level economic attributes		Y	Y	Y
Household characteristics	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
State-specific linear trend				Y

Note: This table estimates the impact of an increase in the maximum weekly UI benefits on the likelihood of enrolling in SNAP for the UI-participating families using the March CPS weights. Results on maximum UI durations, household characteristics and state-level controls are not reported. Standard errors are in parentheses and clustered at the state level. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 6. Effect of Unemployment Insurance on SNAP Benefits, 2nd Part

	(1)	(2)	(3)	(4)
Variable	SNAP benefits			
Maximum weekly UI benefit (in $100)	-243.415**	-210.188**	-234.393***	-177.983*
	(94.206)	(80.573)	(77.467)	(97.852)
Observations	11,159	11,159	11,159	11,159
Mean dependent variable	2877.492	2877.492	2877.492	2877.492
State-level policies			Y	Y
State-level economic attributes		Y	Y	Y
Household characteristics	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
State-specific linear trend				Y

Note: This table estimates the impact of an increase in the maximum weekly UI benefits on the amount of SNAP benefits received by households who participate in both UI and SNAP programs using the March CPS weights. Results on maximum UI durations, household characteristics and state-level controls are not reported. Standard errors are in parentheses and clustered at the state level. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 7. Effect of Unemployment Insurance on Food Insecurity: DDD

	(1)	(2)	(3)	(4)
Variable	Food insecurity status		Food insecurity level	
Maximum weekly UI benefit (in $100)× UI-eligible household	-0.021**	-0.021*	-0.199***	-0.313***
	(0.009)	(0.010)	(0.056)	(0.065)
Maximum weekly UI benefit (in $100)	-0.003**	-0.003	-0.006	0.013
	(0.002)	(0.002)	(0.022)	(0.020)
UI-eligible household	1.416	5.124	-12.674	22.618
	(5.482)	(9.716)	(44.323)	(66.062)
Observations	636,092	636,092	137,225	137,225
Mean dependent variable	0.216	0.216	4.387	4.387
State-level policies	Y	Y	Y	Y
State-level economic attributes	Y	Y	Y	Y
Household characteristics	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
State-specific linear trend		Y		Y

Note: This table estimates the impact of an increase in the maximum weekly UI benefits on household food insecurity using the triple-differences model with the CPS-FSS weights. *UI-eligible household* is coded as 1 if the household is UI-eligible and 0 if the household does not have unemployed members. Results on maximum UI durations, household characteristics, state-level controls, and full pairwise interactions are not reported. Standard errors are in parentheses and clustered at the state level. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 8. Effect of Unemployment Insurance on SNAP Usage: DDD

	(1)	(2)	(3)	(4)
Variable	SNAP enrollment		SNAP benefits	
Maximum weekly UI benefit (in $100)× UI-recipient	-0.010***	-0.010***	-252.114***	-213.968*
	(0.003)	(0.004)	(75.247)	(110.667)
Maximum weekly UI benefit (in $100)	-0.001	-0.001	17.721	35.985
	(0.001)	(0.001)	(21.080)	(33.848)
UI-recipient	-1.383	7.217	20739.617	102271.311
	(1.654)	(5.191)	(41186.555)	(94836.371)
Observations	1,127,326	1,127,326	106,134	106,134
Mean dependent variable	0.094	0.094	3091.869	3091.869
State-level policies	Y	Y	Y	Y
State-level economic attributes	Y	Y	Y	Y
Household characteristics	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
State-specific linear trend		Y		Y

Note: This table estimates the impact of an increase in the maximum weekly UI benefits on household SNAP usage using triple-differences model with the March CPS weights. *UI-recipient* is coded as 1 if the household is an UI recipient and 0 if the household does not have unemployed members. Results on maximum UI durations, household characteristics, state-level controls, and their pairwise interactions are not reported. Standard errors are in parentheses and clustered at the state level. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 9. Effect of Unemployment Insurance on Household UI Income

	(1)	(2)
Variable	Household annual UI income	
Maximum weekly UI benefit (in $100)	495.661***	459.639***
	(80.414)	(70.483)
Observations	78,991	78,991
Mean dependent variable	6346.018	6346.018
State-level welfare policies	Y	Y
State-level economic attributes	Y	Y
Household characteristics	Y	Y
Year FE	Y	Y
State FE	Y	Y
State-specific linear trend		Y

Note: This table estimates the impact of an increase in the maximum weekly UI benefits on the amount of household claimed annual UI benefits with the March CPS weights. Results on maximum UI durations, household characteristics and state-level controls are not reported. Standard errors are in parentheses and clustered at the state level. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 10. Effect of Unemployment Insurance on UI Expenditures

Variable	(1)	(2)	(3)	(4)
	State per-person weekly UI expenditure		State annual UI expenditure (in $1,000)	
Maximum weekly UI benefit (in $100)	32.549***	23.732***	167622.278**	164175.944**
	(9.372)	(8.726)	(68341.866)	(66897.243)
Observations	969	969	969	969
Mean dependent variable	333.923	333.923	896,311.847	896,311.847
State-level welfare policies	Y	Y	Y	Y
State-level economic attributes	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
State-specific linear trend		Y		Y

Note: This table estimates the impact of an increase in the maximum weekly UI benefits on state per-person weekly UI expenditures in the first two columns and state annual UI expenditures in the last two columns. Results on maximum UI durations and state-level controls are not reported. Standard errors are in parentheses and clustered at the state level. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.